Chapter 6

The Metropolis–Hastings Algorithm

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Chapter 5 ended with a Markov chain on a grid of conversion rates whose stationary distribution was the posterior of Chapter 1, built from nothing but ratios of the unnormalized posterior. This chapter turns that idea into the workhorse of Bayesian computation.

The **Metropolis–Hastings (MH)** algorithm builds a Markov chain on a continuous parameter space with any target we can evaluate up to a constant (Sections 6.1 to 6.3).

The **Gibbs sampler** updates one block of parameters at a time from its conditional distribution (Section 6.4). Because the draws are correlated and the chain starts somewhere arbitrary, the chapter closes with the practice of tuning a sampler and checking its output (Sections 6.5 and 6.6).

6.1 From Markov chains to posterior sampling

Example 6.1 (Random-walk Metropolis for the log-odds posterior)**.** The non-conjugate posterior of Example 1.4 has unnormalized density

$$\tilde{\pi}(\eta) = \sigma(\eta)^{12}\big(1-\sigma(\eta)\big)^{188}\,\mathcal{N}\big(\eta;\ \operatorname{logit}(0.05),\ 0.5^2\big).$$

A chain on η can be defined as follows. From the current value η, propose $\eta' = \eta + s\,\varepsilon$ with $\varepsilon \sim \mathcal{N}(0,1)$ and a step size $s = 0.62$; accept the proposal with probability $\min\{1, \tilde{\pi}(\eta')/\tilde{\pi}(\eta)\}$, and otherwise stay at η, as depicted in Figure 6.1.

This is the rule of Example 5.10, but now on the real line. Starting at $\eta = -1$, far in the tail of the posterior, the chain accepts 44.9% of the proposals. After discarding the first 1 000 of 20 000 iterations, the draws give

$$\mathbb{E}[\theta \mid \mathbf{y}] \approx 0.0577 \pm 0.0005, \qquad \Pr(\theta > 0.05 \mid \mathbf{y}) \approx 0.690 \pm 0.013,$$

against the quadrature values 0.0576 and 0.6845 of Example 1.4 (Figure 6.2). The margins use the ESS of Theorem 5.2, 3 545 and 4 568 out of 19 000 draws. The naive standard error of the posterior mean, 0.00010, would understate the correct one, 0.00024, by more than half.

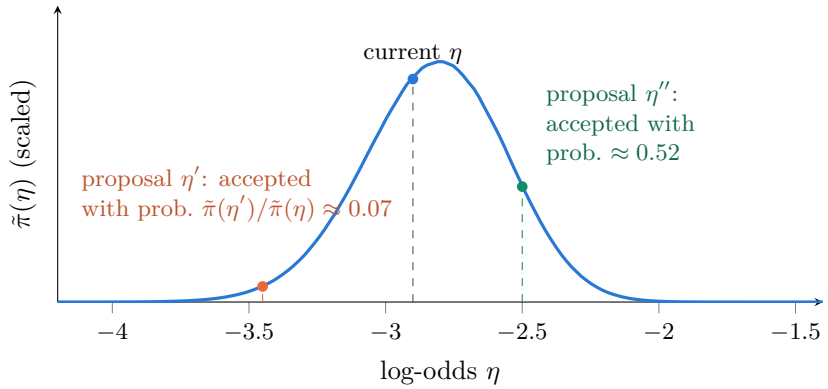


Figure 6.1: One Metropolis step (Example 6.1). From the current value η, a proposal is accepted with probability equal to the ratio of the heights of the target, capped at one: rarely in the tail (0.07), often near the center (0.52).

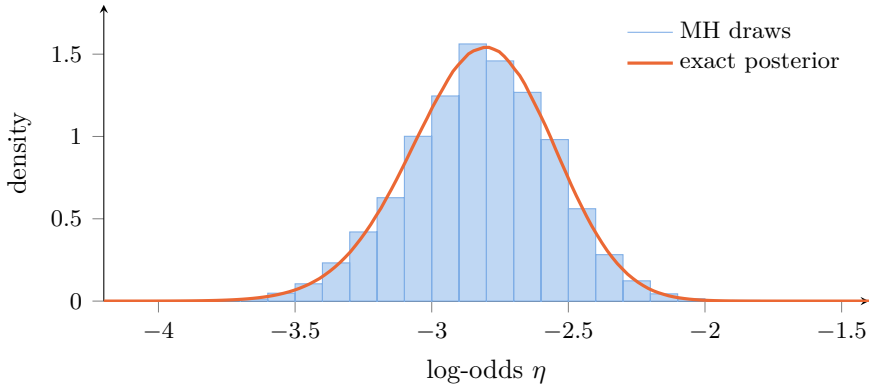


Figure 6.2: Histogram of 19 000 random-walk Metropolis draws of the log-odds η and the exact posterior density, normalized by quadrature (Example 6.1).

The principle is general. Given a target density π on $\Theta \subseteq \mathbb{R}^d$, known up to a constant, we want a transition kernel K such that (i) π is stationary for K, (ii) the chain is irreducible and aperiodic, so that it converges to π from any start (Theorem 5.1), and (iii) it moves fast, so that the integrated autocorrelation time τ is small and the ergodic averages of Theorem 5.2 are precise. Detailed balance (Proposition 5.2) takes care of (i) with ratios of π only; the choice of proposal decides (ii) and (iii). This is **MCMC**: Monte Carlo integration (Chapter 3) with draws produced by a Markov chain.

6.2 The Metropolis–Hastings recipe

The MH algorithm (Metropolis et al., 1953; Hastings, 1970) allows any proposal density $q(\theta' \mid \theta)$ that we can sample and evaluate. The factor $q(\theta \mid \theta')/q(\theta' \mid \theta)$ in the acceptance probability corrects for proposals that move more easily in one direction than in the other. When q is symmetric, $q(\theta' \mid \theta) = q(\theta \mid \theta')$, the correction cancels and we recover the **Metropolis** acceptance probability $\min\{1, \tilde\pi(\theta')/\tilde\pi(\theta)\}$ of Example 6.1.

Theorem 6.1 (The Metropolis–Hastings kernel leaves π invariant)**.** *The Markov chain produced by Algorithm 4 satisfies detailed balance with respect to $\pi = \tilde\pi/Z$, hence π is a stationary distribution. If, in addition, the chain is irreducible and aperiodic (for example, if $q(\theta' \mid \theta) > 0$ wherever $\pi(\theta') > 0$), the distribution of $\theta^{(t)}$ converges to π and the ergodic theorem applies.*

Algorithm 4: Metropolis–Hastings

Input: unnormalized target $\tilde{\pi}$, proposal density $q(\cdot \mid \theta)$, starting value $\theta^{(0)}$, number of iterations N

for $t = 1, \dots, N$ **do**
  draw a proposal $\theta' \sim q(\cdot \mid \theta^{(t-1)})$ and $U \sim \mathcal{U}(0,1)$;
  compute $\alpha = \min\left\{1,\ \dfrac{\tilde{\pi}(\theta')\, q(\theta^{(t-1)} \mid \theta')}{\tilde{\pi}(\theta^{(t-1)})\, q(\theta' \mid \theta^{(t-1)})}\right\}$;
  if $U \le \alpha$ **then**
    $\theta^{(t)} \leftarrow \theta'$; `// accept`
  else
    $\theta^{(t)} \leftarrow \theta^{(t-1)}$; `// reject: stay`

Proof. For $\theta' \neq \theta$, the chain moves from θ to θ' with density $q(\theta' \mid \theta)\,\alpha(\theta, \theta')$. Write $r = \pi(\theta')q(\theta \mid \theta')/\{\pi(\theta)q(\theta' \mid \theta)\}$. If $r \le 1$, then $\alpha(\theta, \theta') = r$ and $\alpha(\theta', \theta) = 1$, so

$$\pi(\theta)\, q(\theta' \mid \theta)\, \alpha(\theta, \theta') = \pi(\theta')\, q(\theta \mid \theta') = \pi(\theta')\, q(\theta \mid \theta')\, \alpha(\theta', \theta);$$

the case $r > 1$ is symmetric. The rejected proposals only add mass at the current point, which does not affect detailed balance. Stationarity follows as in Proposition 5.2, with sums replaced by integrals; convergence follows from the general-state-space version of Theorem 5.1 (Robert and Casella, 2004). □

Three remarks make the algorithm practical.

- *Only ratios of $\tilde{\pi}$ are needed.* The evidence cancels, as anticipated in Remark 1.5. Ratios are computed as differences of log densities, accepted when $\log U \le \log \tilde{\pi}(\theta') - \log \tilde{\pi}(\theta)$, to avoid the underflow of Remark 1.3.
- *Rejections are part of the output.* When a proposal is rejected, the current value is recorded again; dropping repeated values would bias the averages.
- *Constrained parameters.* A random walk is easiest on $\mathbb{R}^d$. Positive parameters are sampled on the log scale and probabilities on the logit scale, and the target then includes the Jacobian of the transformation (Proposition 2.1), as in Lab 6.

Example 6.2 (Three channel coefficients)**.** The two-channel model of Lab 3 has three coefficients (intercept, TV, search), with 2 000 visitors. A multivariate random walk proposes $\beta' = \beta + L\varepsilon$, with $\varepsilon \sim \mathcal{N}(0, I_3)$ and $LL^\top$ a proposal covariance. A short pilot run with a small isotropic step (3 000 iterations, acceptance 0.55) provides an estimate $\hat{\Sigma}$ of the posterior covariance; the main run uses $(2.38^2/d)\,\hat{\Sigma}$ with $d = 3$ (Section 6.5) and accepts 28% of the proposals. From 28 000 draws: posterior means -3.259, 0.128 and 0.984 with standard deviations 0.155, 0.194 and 0.188, and $\Pr(\beta_{\text{search}} > \beta_{\text{TV}} \mid \mathbf{y}) = 0.9993$. These agree with the importance sampling estimates of Lab 3 (-3.258, 0.131, 0.982; 0.999), obtained with a completely different algorithm. Each coefficient has an ESS of about 2 700–3 000.

6.3 Random-walk and independence samplers

Two families of proposals cover most practice.

Random walk. $\theta' = \theta + s\,\varepsilon$ (or $\theta + L\varepsilon$) with a symmetric distribution for ε. The proposal explores locally around the current point and needs no prior knowledge of where the posterior lives, but the step size must be tuned (Section 6.5).

Independence sampler. $\theta' \sim g$, a fixed distribution that does not depend on the current point. Then $q(\theta' \mid \theta) = g(\theta')$ and the acceptance probability becomes

$$\alpha(\theta, \theta') = \min\left\{1, \frac{\tilde{\pi}(\theta')/g(\theta')}{\tilde{\pi}(\theta)/g(\theta)}\right\} = \min\left\{1, \frac{w(\theta')}{w(\theta)}\right\},$$

the ratio of the importance weights of Section 3.4. A good importance proposal makes a good independence sampler; a bad one makes a sticky chain that stays for long periods at points with very large weights.

Example 6.3 (Proposals for the log-odds posterior)**.** With 20 000 iterations on the log-odds posterior:

Sampler	Acceptance	$\mathbb{E}[\theta \mid \mathbf{y}]$	ESS of θ
random walk, $s = 0.62$	0.449	0.0577	3 545
independence, Laplace Student-t_4	0.906	0.0576	15 876
independence, $\mathcal{N}(\hat{\eta}, 0.05^2)$	0.442	0.0586	82

The Laplace Student-t_4 proposal of Example 3.4 is so close to the posterior that 91% of the proposals are accepted, and the draws are almost independent. The proposal that is too narrow accepts about as often as the random walk, but its rare excursions into the tails are kept for many iterations, and 19 000 draws are worth 82 independent ones. In one dimension a well-matched independence proposal wins; in higher dimensions, finding such a proposal is as hard as the original problem (Example 3.4), and random walks, which need only local information, are the default.

6.4 The Gibbs sampler

Example 6.4 (The two segments, with labels as latent variables)**.** Return to the two customer segments of unequal size of Example 4.5: shares 0.3 and 0.7, $\sigma = 10$, prior $\mathcal{N}(65, 30^2)$ on each mean, 100 simulated purchase amounts. Add to the unknowns the segment z_i of each buyer. Given the means, each label is a coin flip with the responsibility r_i of the EM E-step (Example 4.7); given the labels, each mean has a conjugate Normal posterior:

$$z_i \mid \mu \sim \text{Bernoulli}(r_i), \qquad \mu_k \mid z \sim \mathcal{N}\Big(v_k\Big(\frac{65}{30^2} + \frac{\sum_{i:z_i=k} y_i}{\sigma^2}\Big),\ v_k\Big), \quad v_k = \Big(\frac{1}{30^2} + \frac{n_k}{\sigma^2}\Big)^{-1},$$

with n_k the number of buyers currently in segment k. Alternating these two draws gives a Markov chain on (z, μ_1, μ_2). Started near the global mode $(40, 90)$, 5 000 iterations give posterior means $(38.6, 91.2)$; started near the local mode $(90, 40)$, they give $(91.6, 39.3)$, and the chain never once visits the other mode (Figure 6.3). Each run looks converged; together they disagree. We return to this in Section 6.6.

Definition 6.1 (Gibbs sampler)**.** Split the unknowns into blocks, $\theta = (\theta_1, \ldots, \theta_B)$. The **full conditional** of block b is its distribution given all the other blocks and the data, $\pi(\theta_b \mid \theta_{-b}, \mathbf{y}) \propto \pi(\theta \mid \mathbf{y})$. One **systematic-scan** iteration of the Gibbs sampler (Geman and Geman, 1984; Gelfand and Smith, 1990) draws in turn

$$\theta_1^{(t)} \sim \pi(\theta_1 \mid \theta_2^{(t-1)}, \ldots, \theta_B^{(t-1)}, \mathbf{y}),\ \ldots,\ \theta_B^{(t)} \sim \pi(\theta_B \mid \theta_1^{(t)}, \ldots, \theta_{B-1}^{(t)}, \mathbf{y}).$$

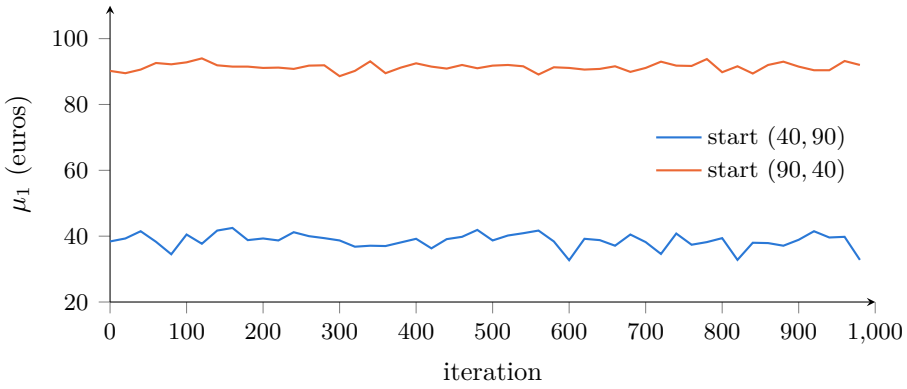


Figure 6.3: Gibbs sampler for the two segment means (Example 6.4): traces of μ_1 for chains started in the two modes. Neither chain ever crosses to the other mode.

Proposition 6.1 (Gibbs is Metropolis–Hastings with acceptance one)**.** *Updating block b by a draw from its full conditional is an MH step with proposal $q(\theta' \mid \theta) = \pi(\theta'_b \mid \theta_{-b}, \mathbf{y})$, $\theta'_{-b} = \theta_{-b}$, whose acceptance probability is always one. Hence, each update and the whole scan leave the posterior invariant.*

Proof. With $\theta'_{-b} = \theta_{-b}$, factor $\pi(\theta) = \pi(\theta_{-b})\,\pi(\theta_b \mid \theta_{-b})$. The MH ratio is

$$\frac{\pi(\theta')\,q(\theta \mid \theta')}{\pi(\theta)\,q(\theta' \mid \theta)} = \frac{\pi(\theta_{-b})\,\pi(\theta'_b \mid \theta_{-b}) \cdot \pi(\theta_b \mid \theta_{-b})}{\pi(\theta_{-b})\,\pi(\theta_b \mid \theta_{-b}) \cdot \pi(\theta'_b \mid \theta_{-b})} = 1.$$

A composition of kernels that each leave π invariant leaves π invariant. □

The Gibbs sampler needs no tuning, but it needs full conditionals that are easy to sample, which conjugate structure provides; any block without a standard full conditional can be updated with an MH step instead (**Metropolis-within-Gibbs**). Its weak point is visible in Example 6.4: when blocks are strongly dependent (the labels are almost determined by the means, and vice versa), each conditional move is small, and the chain crawls or stays trapped.

Example 6.5 (Daily orders in ten regions)**.** A new product was launched in ten regions at different dates: region j has been observed for t_j days (1 to 60) and received y_j orders. The daily rates λ_j differ between regions but are drawn from a common distribution, whose scale is itself unknown:

$$y_j \mid \lambda_j \sim \text{Poisson}(\lambda_j t_j), \qquad \lambda_j \mid \beta \overset{\text{iid}}{\sim} \text{Gamma}(\text{shape } 10, \text{rate } \beta),$$
$$\beta \sim \text{Gamma}(\text{shape } 1, \text{rate } 1).$$

Such a **hierarchical model** is a Bayesian network with one more layer than Figure 1.3. Conjugacy gives both full conditionals:

$$\lambda_j \mid \beta, \mathbf{y} \sim \text{Gamma}(10 + y_j,\ \beta + t_j), \qquad \beta \mid \lambda \sim \text{Gamma}\Big(1 + 10J,\ 1 + \sum_j \lambda_j\Big),$$

with $J = 10$ regions (second arguments are rates). From 20 000 Gibbs iterations, the common mean rate is $10/\mathbb{E}[\beta \mid \mathbf{y}] \approx 12.2$ orders per day. A region observed for a single day with 18 orders gets a posterior mean of 15.4 (raw rate 18, true rate 10.9); a region observed for 60 days barely moves (7.87 → 7.93). This **shrinkage** borrows strength across regions: the average absolute error against the true rates falls from 1.79 (raw rates) to 1.31 (posterior means). The ESS of β is 12 794 out of 19 000 draws: here the blocks are only weakly dependent, and Gibbs mixes well.

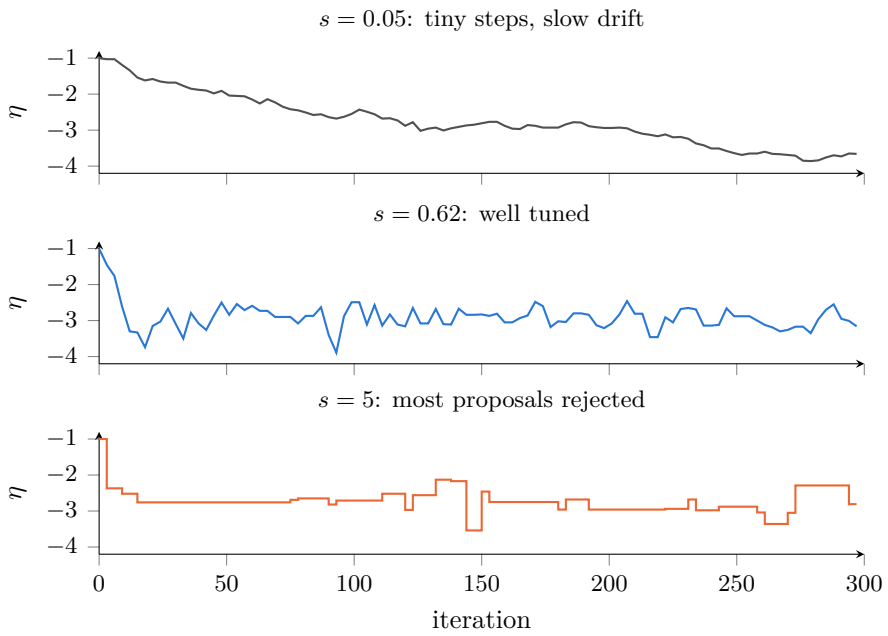


Figure 6.4: First 300 iterations of random-walk Metropolis from $\eta = -1$ with three step sizes (Example 6.6). Too small: a slow drift; well tuned: rapid exploration; too large: long flat stretches of rejections.

6.5 Tuning and mixing

Example 6.6 (Choosing the step size)**.** Run the random-walk sampler of Example 6.1 with five step sizes (20 000 iterations after a burn-in of 1 000):

Step s	**Acceptance rate**	**ESS per** 10 000 draws	lag-1 **autocorrelation**
0.05	0.937	58	0.983
0.20	0.766	813	0.832
0.62	0.448	2 291	0.620
1.50	0.215	1 515	0.738
5.00	0.065	392	0.914

With tiny steps, nearly every proposal is accepted, but the chain crawls; with huge steps, nearly every proposal is rejected and the chain sticks (Figure 6.4). The best step accepts slightly fewer than half of the proposals and is 40 times more efficient than the smallest one.

Theory backs this rule of thumb. For a target that is a product of d identical one-dimensional distributions and a Gaussian random walk, Roberts et al. (1997) show that, as d grows, the efficiency is maximized when the proposal standard deviation is about $2.38/\sqrt{d}$ times the posterior standard deviation in each direction, which yields an acceptance rate of about 0.234; in one dimension the optimal acceptance rate is about 0.44. When the posterior is correlated, the proposal should follow its shape: the covariance $(2.38^2/d)\,\hat{\Sigma}$, with $\hat{\Sigma}$ estimated from a pilot run, as in Example 6.2. Adaptive samplers update $\hat{\Sigma}$ on the fly while preserving convergence (Haario et al., 2001). In practice:

- start from a reasonable point (the MAP estimate, Chapter 4) and a proposal scale from the curvature there (the Laplace approximation);
- run a pilot, estimate the posterior covariance, rescale by $2.38^2/d$, and aim for an acceptance rate between 0.2 and 0.5;

- reparametrize to reduce correlations and constraints: log or logit scales, centering, and rescaling of covariates.

6.6 Convergence diagnostics

An MCMC run raises two questions: has the chain forgotten its starting point (**convergence**), and how precise are the averages (**mixing**)? No diagnostic proves convergence; the following tools can only detect failures, and they are used together.

- *Trace plots* of each parameter against the iteration (Figures 6.3 and 6.4): a converged, well-mixing chain looks like a stationary "hairy caterpillar"; trends, long flat stretches, or jumps between levels are warnings.
- *Burn-in*: discard the first iterations, during which the chain travels from its starting point to the bulk of the posterior.
- *Autocorrelations* ρ_k and the ESS $= N/\tau$ of Theorem 5.2, with Monte Carlo standard errors $\hat{\sigma}/\sqrt{\text{ESS}}$. For the log-odds chain, the autocorrelations at lags 1, 5 and 10 are 0.62, 0.11 and 0.00 with $s = 0.62$, but 0.98, 0.92 and 0.84 with $s = 0.05$.
- *Several chains* from dispersed starting points, compared with $\hat{R}$.

Definition 6.2 (Split-$\hat{R}$)**.** Run m chains of length $2n$ and split each in two halves, giving $2m$ sequences of length n. Let W be the average of the within-sequence variances and B/n the variance of the sequence means. The **potential scale reduction factor** is

$$\hat{R} = \sqrt{\frac{\frac{n-1}{n}W + \frac{B}{n}}{W}}.$$

If all sequences sample the same distribution, the between- and within-sequence variances agree and $\hat{R} \approx 1$. Values above 1.01 indicate that the chains have not mixed (Gelman and Rubin, 1992; Vehtari et al., 2021); splitting each chain also detects a chain that is still drifting.

Example 6.7 (Diagnostics in action)**.** For the log-odds posterior, four random-walk chains started at $\eta = -6, -4, -1.5$ and 0.5, with 4 500 draws each after a burn-in of 500, give $\hat{R} = 1.0006$ and a total ESS of 3 826. Using only their first 100 iterations, without burn-in, gives $\hat{R} = 1.18$: the chains have not yet forgotten their starting points. For the two segments of Example 6.4, four Gibbs chains started near each of the two modes give $\hat{R} = 17.8$ for μ_1, with chain means 38.6 and 91.5: a clear alarm, raised although each chain on its own looked converged.

The converse is a warning: had all four chains started near the global mode, $\hat{R}$ would have been close to one and the local mode would have gone unnoticed. Diagnostics detect what the chains have seen; dispersed starting points, and an understanding of the model (here, the possibility of label switching), are part of the diagnosis.

Summary

Takeaways

- MH proposes $\theta' \sim q(\cdot \mid \theta)$ and accepts with probability $\min\{1, \tilde{\pi}(\theta')q(\theta \mid \theta')/[\tilde{\pi}(\theta)q(\theta' \mid \theta)]\}$; detailed balance makes the posterior

stationary, and only ratios of the unnormalized posterior are needed.

- Random-walk proposals need only local information but must be tuned; independence proposals work when they resemble the posterior, with acceptance governed by importance weights.
- Gibbs samples each block from its full conditional (an MH step with acceptance one); it needs no tuning but mixes poorly when blocks are strongly dependent.
- Tune the step to an acceptance rate of about 0.2–0.5, with a proposal covariance $(2.38^2/d)\hat{\Sigma}$ from a pilot run.
- Check every run: traces, burn-in, autocorrelation, ESS and split-$\hat{R}$ from several chains with dispersed starts; diagnostics detect failures but never prove convergence.

Further reading

Chapters 7 to 12 of Robert and Casella (2004) treat the MH algorithm, the Gibbs sampler, and convergence diagnostics in depth. Chapters 5 and 6 of Gamerman and Lopes (2006) give a Bayesian-oriented account. Chapters 11 and 12 of Gelman et al. (2013) present MCMC practice, including $\hat{R}$ and the effective sample size, and Brooks et al. (2011) collects reviews on most aspects of MCMC. The original papers are Metropolis et al. (1953) and Hastings (1970). The Gibbs sampler was introduced by Geman and Geman (1984) and popularized for Bayesian computation by Gelfand and Smith (1990). Optimal scaling is due to Roberts et al. (1997), and the improved $\hat{R}$ is due to Vehtari et al. (2021).